

Implied Quarterly Forecasts from Annual Projections in the ECB Survey of Professional Forecasters*

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Abstract

This paper recovers quarterly fixed-horizon forecasts of GDP growth from annual fixed-event projections in the European Central Bank (ECB) Survey of Professional Forecasters (SPF). Based on an approximation of annual projections by quarterly growth rates, we use a Kalman filter and smoother to obtain SPF-consistent forecasts of quarterly GDP growth. Using real-time GDP data, we apply this approach at both the consensus and individual level, resulting in a cross-section of implied forecasts without look-ahead bias. The implied consensus forecasts outperform standard time-series benchmarks at short horizons of up to around three quarters ahead, but appear uninformative at longer horizons. Consistent with earlier findings, the implied consensus forecasts exhibit underreaction, while the individual SPF-consistent forecasts show signs of overreaction.

Keywords: Fixed-horizon forecasts, Fixed-event forecasts, State-space models, Forecast evaluation, Forecast rationality

JEL Classification: C53, D84, E37

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1 Introduction

Quarterly forecasts from professional forecasters receive considerable attention both in academia and among practitioners. For example, the U.S. Survey of Professional Forecasters provides quarterly projections for key macroeconomic variables that are closely monitored by policymakers and the public. These forecasts are also valuable inputs for models used to assess the current macroeconomic outlook. Moreover, empirical evidence suggests that survey forecasts for GDP growth are most informative at short horizons, while forecast performance deteriorates at longer horizons, highlighting the relevance of quarterly expectations. In contrast, the European Central Bank (ECB) Survey of Professional Forecasters (SPF) reports projections for euro area GDP growth only at the annual level. As a result, quarterly expectations about euro area output growth cannot be directly observed in the survey, limiting the analysis of short-term macroeconomic expectations.

In this paper, we follow [Hepenstrick and Blunier \(2022\)](#) and recover quarterly forecasts that are consistent with annual ECB SPF projections. Based on a state-space representation approximating yearly GDP growth by quarterly growth rates, this approach links annual fixed-event projections to implied quarterly forecasts. The resulting state-space model can be estimated via Kalman filter-based quasi-maximum-likelihood. The smoothed path of quarterly growth rates implied by the filter represents SPF-consistent quarterly forecasts. Using vintage data of euro area GDP, we apply this methodology in real time and construct filtered ECB SPF forecasts without look-ahead bias. We use the filter both for ECB consensus projections and for individual forecasters, allowing us to obtain a panel of SPF-consistent quarterly forecasts. We evaluate these implied quarterly projections in real time and study deviations from rationality as well as heterogeneity across forecasters.

A large body of literature studies forecasts from surveys of professionals. For an extensive review of research using both the U.S. and ECB SPF, see [Clements et al. \(2023\)](#). This strand of literature examines forecast accuracy ([Stark, 2010](#); [Faust and Wright, 2013](#)) and deviations from rationality ([Coibion and Gorodnichenko, 2015](#); [Bordalo et al., 2020](#)), as well as measures of uncertainty ([Abel et al., 2016](#); [Rich and Tracy, 2021](#)) and disagreement

among professional forecasters ([Glas, 2020](#); [Clements, 2022](#)). Moreover, SPF forecasts are used as inputs in macroeconomic risk assessment and to refine density forecasts ([Adams et al., 2021](#); [Krüger et al., 2017](#); [Schick, 2024](#)). These studies typically analyze forecasts at the quarterly frequency, whereas the ECB SPF provides GDP projections only at the annual frequency.

A related literature studies how fixed-event forecasts can be converted into fixed-horizon forecasts. Fixed-event forecasts, such as GDP growth for a given calendar year, exhibit a seasonal pattern because the implied forecast horizon decreases as the year progresses. Many empirical applications, however, require forecasts at fixed horizons—for example, GDP growth one year ahead. Existing approaches typically construct such forecasts using ad hoc weighting schemes ([Dovern et al., 2012](#); [Knüppel and Vladu, 2025](#)) or by combining fixed-event densities to obtain correctly calibrated fixed-horizon density forecasts ([Ganics et al., 2024](#)). While these approaches focus on aligning forecast horizons, they do not address situations in which projections are reported only at the annual level, even though many applications require forecasts at the quarterly frequency. To recover higher-frequency forecasts, we follow [Hepenstrick and Blunier \(2022\)](#), who formulate a state-space representation based on approximating annual GDP growth by quarterly growth rates ([Tödter, 2010](#)). This approach allows us to recover the implied quarterly forecasts underlying the annual fixed-event projections.

This paper makes several contributions to the literature on survey forecasts. First, it constructs quarterly forecasts for the ECB SPF. Following [Hepenstrick and Blunier \(2022\)](#), we use annual fixed-event consensus projections to recover SPF-consistent quarterly forecasts without look-ahead bias. Second, we provide a real-time forecast evaluation by comparing the accuracy of the filtered consensus forecasts with common benchmark models. Third, we apply the state-space approach to individual forecasters and construct a panel of filtered quarterly fixed-horizon forecasts. This allows us to study deviations from rationality and disagreement in the cross-section of implied quarterly SPF forecasts.

We find that the filter approach yields implied forecasts with properties broadly

consistent with existing evidence on quarterly survey projections. A realistic forecast evaluation shows that the implied quarterly SPF forecasts are accurate at short horizons and significantly outperform univariate time-series models for horizons up to two quarters ahead. At longer horizons, however, their predictive accuracy is no better than a backward-looking sample mean, indicating limited informational content beyond simple benchmarks. The implied forecasts are approximately unbiased and efficient at very short horizons but display systematic deviations from rationality as the forecast horizon increases. Consistent with earlier evidence, we find underreaction in consensus forecasts and overreaction at the individual forecaster level (Coibion and Gorodnichenko, 2015; Bordalo et al., 2020). Finally, the panel of filtered forecasts reveals substantial heterogeneity across forecasters: disagreement increases with the forecast horizon and varies over time, particularly following recessions (Patton and Timmermann, 2010; Conrad and Lahiri, 2025).

The paper is organized as follows. Section 2 provides a brief review of the related literature. Section 3 describes the SPF and the real-time data on GDP growth. The state-space representation as well as the estimation and filtering procedure are outlined in Section 4. Section 5 explains the forecast evaluation, and Section 6 discusses the empirical results. Section 7 concludes the paper.

2 Related literature

A large literature studies survey expectations and their role in macroeconomic forecasting. In particular, many studies examine the accuracy of survey forecasts and find that their predictive performance is horizon dependent. Consensus survey forecasts are often found to outperform time-series models at short horizons (Stark, 2010; Faust and Wright, 2013; Giannone et al., 2008; Bańbura et al., 2013; Bańbura et al., 2021). This superior performance is likely attributable to the aggregation of forecasts across a large cross-section of survey respondents, each potentially incorporating a rich information set. In addition, expert judgment allows forecasters to quickly adapt to structural breaks and

large economic shocks, such as the global financial crisis ([Ang et al., 2007](#)). Consequently, the informational content of survey expectations is particularly strong at short horizons. Our paper contributes to this literature by recovering implied quarterly forecasts from annual projections, which enables us to study the forecast accuracy of the ECB SPF at short horizons.

Beyond their forecast accuracy, survey expectations are also widely used as inputs in macroeconomic forecasting models. Several studies show that incorporating survey expectations can improve point and density forecasts of key macroeconomic variables ([Krüger et al., 2017](#); [Adams et al., 2021](#); [Bańbura et al., 2021](#)). Survey information has also been used in the construction of macroeconomic risk measures and prediction intervals ([Krüger and Plett, 2024](#); [Schick, 2024](#)), as well as in scenario analysis and stress-testing exercises ([Schick and Opschoor, 2025](#)). These applications highlight the informational content of survey expectations for macroeconomic forecasting. Our paper contributes by providing higher-frequency survey expectations for the euro area that may serve as inputs in such applications.

Another strand of the literature studies the rationality of survey expectations. Empirical tests typically examine whether survey forecasts are unbiased and efficient ([Mincer and Zarnowitz, 1969](#); [Nordhaus, 1987](#)). Survey forecasts are often found to be biased, although the degree of bias depends on the forecast horizon ([Juodis and Kučinskas, 2023](#); [Heinisch, 2025](#)). Moreover, forecasts tend to be more strongly biased at the individual than at the consensus level ([Zarnowitz and Lambros, 1987](#)). While survey forecasts often perform well at short horizons, a growing literature documents systematic deviations from full-information rational expectations. In particular, forecasts are found to display patterns of underreaction or overreaction to new information, depending on the level of aggregation ([Kohlhas and Walther, 2021](#); [Coibion and Gorodnichenko, 2015](#); [Bordalo et al., 2020](#); [Bürgi and Ortiz, 2026](#)). Such deviations may arise from informational rigidities or behavioral biases in expectation formation. Alternative explanations emphasize the role of asymmetric loss functions of forecasters ([Capistrán and Timmermann, 2009](#); [Elliott et al., 2008](#)).

Closely related to rationality is disagreement and heterogeneity among professional forecasters. Disagreement is typically measured as the cross-sectional dispersion of individual forecasts (Abel et al., 2016; Glas, 2020). Empirical evidence shows that disagreement tends to increase with the forecast horizon (Patton and Timmermann, 2010; Lahiri and Sheng, 2010; Conrad and Lahiri, 2025). In addition, disagreement varies over time and is often particularly pronounced during periods of economic downturns (Patton and Timmermann, 2010; Doornik et al., 2012). These findings point to substantial heterogeneity in expectations across individual forecasters. Our paper contributes to these strands of the literature by constructing a panel of implied quarterly SPF forecasts, which allows us to examine rationality and disagreement at both the individual and the consensus level.

Our paper is closely related to the literature on fixed-event and fixed-horizon forecasts. Fixed-event forecasts imply a seasonal horizon structure because the effective forecast horizon shortens as the year progresses. Early approaches convert fixed-event forecasts into fixed-horizon forecasts using ad hoc weighting schemes (see, e.g., Doornik et al., 2012; Siklos, 2013; Hubert, 2015). More recently, Knüppel and Vladu (2025) proposes an optimal approximation that minimizes the mean-squared approximation error of the resulting forecasts. Related work develops methods to transform fixed-event density forecasts into fixed-horizon densities (Ganics et al., 2024).

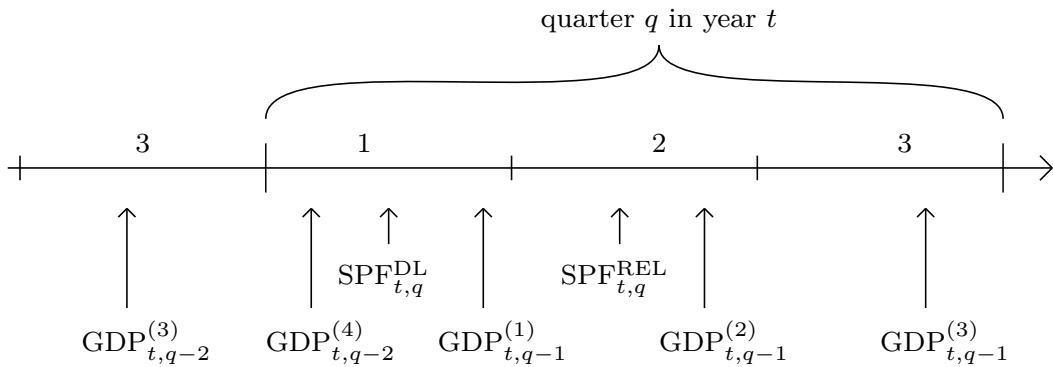
A related literature relies on state-space models to recover higher-frequency expectations from lower-frequency survey projections. Several papers apply such methods to estimate term structures of inflation expectations (Kozicki and Tinsley, 2012; Aruoba, 2020; Mertens and Nason, 2020). Related contributions construct term structures of survey uncertainty (Clark et al., 2025), inflation densities (Grishchenko et al., 2019), and disagreement among U.S. professional forecasters (Crump et al., 2023). Most closely related to our paper, Hepenstrick and Blunier (2022) use a state-space representation based on approximating annual growth rates by quarterly growth rates to recover implied quarterly forecasts from annual survey projections. We apply this approach to the ECB SPF and evaluate the resulting quarterly expectations in real time.

3 Data and real-time information set

The ECB-SPF is a quarterly survey that began in 1999 and provides forecasts for key macroeconomic variables in the euro area. Panelists come from both financial and non-financial institutions and are predominantly located within the euro area (Clements et al., 2023). Over the sample period, the number of respondents ranges from 47 to 72, with a median of 57 participants per round. For real GDP growth, panelists report annual fixed-event forecasts for the current and the next two calendar years, as well as quarterly year-on-year fixed-horizon forecasts at one- and two-year-ahead horizons. For each forecast, the survey provides both point and density predictions.

Figure 1 sketches the deadline and release timing of the survey, along with the releases and revisions of the most recent GDP figures. The deadline to submit forecasts usually falls in the first month of a quarter, prior to the publication of the first available estimate of GDP for the previous quarter, $GDP_{q-1}^{(1)}$. In more recent years, GDP figures are revised monthly, although revisions occurred at a lower frequency prior to 2016. Panelists always know at least the first release of quarter $t - 2$ growth rates and, in more recent years, occasionally even the third revised value $GDP_{t,q-2}^{(4)}$.

Figure 1: Release calendar



Notes: This graph sketches the release and revision dates of GDP together with the deadline for submission and the release dates of the Survey of Professional Forecasters, i.e., $SPF_{t,q}^{DL}$ and $SPF_{t,q}^{REL}$, respectively. For instance, $GDP_{t,q-1}^{(1)}$ denotes the first release of the previous quarter's GDP figures. The numbers on the timeline indicate the months within each quarter.

In this paper, we focus on the point forecasts of the SPF. The fixed-event annual GDP

growth projection of panelist i for year $t + j$ reported in the survey conducted in quarter q of year t is

$$\text{SPF}_{t+j|t,q}^{FE,i} = 100 \left(\frac{\text{GDP}_{t+j|t,q}^{A,i}}{\text{GDP}_{t+j-1|t,q}^{A,i}} - 1 \right), \quad (1)$$

where $\text{GDP}_{t+j|t,q}^{A,i}$ denotes panelist i 's forecast of annual GDP in year $t + j$, conditional on the information available in quarter q of year t . Here, $j = 0$ corresponds to the current year and $j = 1$ to the next year.

In addition, the SPF collects year-on-year fixed-horizon forecasts. The one-year-ahead forecast starts from the most recently released quarterly GDP growth:

$$\text{SPF}_{q+2|t,q}^{FH,i} = 100 \left(\frac{\text{GDP}_{q+2|t,q}^i}{\text{GDP}_{q-2|t,q}} - 1 \right), \quad (2)$$

where $\text{GDP}_{q-2|t,q}$ denotes the latest released quarterly GDP growth available at the time of the survey, i.e., $\text{GDP}_{t,q-2}^{(3)}$ or $\text{GDP}_{t,q-2}^{(4)}$ in Figure 1, and $\text{GDP}_{q+2|t,q}^i$ is panelist i 's four-quarter-ahead forecast reported in the survey conducted in quarter q of year t .

For example, in the first quarterly survey of a year ($q = 1$), the most recent vintage contains GDP growth up to $q = 3$ of the previous year ($t - 1$). Hence, the four-quarter-ahead forecast $\text{SPF}_{q+2|t,q}^{FH,i}$ represents the year-on-year GDP growth forecast for the third quarter of year t .

As the consensus forecast, we use the median across individual projections:

$$\begin{aligned} \text{SPF}_{t+j|t,q}^{FE} &= \text{median}_i \left(\text{SPF}_{t+j|t,q}^{FE,i} \right), \\ \text{SPF}_{q+2|t,q}^{FH} &= \text{median}_i \left(\text{SPF}_{q+2|t,q}^{FH,i} \right), \end{aligned}$$

where the median is taken across panelists i .¹ Because our analysis focuses on short-term horizons, we retain only three projections per survey round: the current- and next-year fixed-event forecasts and the one-year-ahead fixed-horizon forecast.

¹Results are virtually identical when using the mean as the consensus forecast.

Real-time quarterly GDP growth rates are constructed using the GDP vintage available at each SPF survey deadline. Since time is indexed by year and quarter, we define the quarterly lag operator L on a quarterly variable $x_{t,q}$ as

$$L(x_{t,q}) = \begin{cases} x_{t,q-1} & \text{if } q > 1 \\ x_{t-1,4} & \text{if } q = 4. \end{cases} \quad (3)$$

Annualized quarterly growth rates are defined as

$$y_{t,q} = 100 \left(\left(\frac{\text{GDP}_{t,q}}{L(\text{GDP}_{t,q})} \right)^4 - 1 \right),$$

where $\text{GDP}_{t,q}$ denotes real GDP in quarter q of year t taken from the latest vintage available at the corresponding SPF survey deadline. For each survey round, we construct the full history of quarterly growth rates available in that vintage at the survey deadline.

For the SPF, the submission deadlines and release dates of each survey round are publicly available. To reconstruct the information set of survey respondents as closely as possible, we align these deadlines with real-time GDP vintages from the ECB Euro Area Real-Time Database. Exact release dates for each vintage are available from the third quarter of 2014 onward.

Prior to 2014, only the release month of each vintage is known. To approximate the information available to panelists before 2014, we make the conservative assumption that the monthly vintages were available at the end of the month. Therefore, some revisions may not be taken into account. However, at the SPF deadline panelists know at least the first release of GDP for quarter $t - 2$. Thus, if our conservative approach matches the SPF with a vintage from the previous month that does not include GDP for quarter $t - 2$, we infer that the vintage released in the same month as the survey deadline was already available when respondents submitted their forecasts. This occurs only once, in the second survey round of 2002.

4 Methodology

Since the ECB-SPF only provides annual projections, this section details how we recover quarterly h -step-ahead forecasts that are consistent with the consensus survey responses. To this end, we first outline an approximation that links annual GDP growth to annualized quarterly growth rates. This approximation admits a state-space representation, which allows us to estimate quarterly SPF-consistent forecasts using a Kalman filter-based quasi maximum likelihood (QML) approach, following [Hepenstrick and Blunier \(2022\)](#).

4.1 State-space representation

In the main part of the analysis, we only use fixed-event forecasts. In the robustness section, we show how this state-space representation can be extended to incorporate fixed-horizon forecasts. To link quarterly to annual GDP growth, we use the following approximation:

$$y_t = \frac{1}{16}(x_{t,4} + 2x_{t,3} + 3x_{t,2} + 4x_{t,1} + 3x_{t-1,4} + 2x_{t-1,3} + x_{t-1,2}) + e_t, \quad (4)$$

where y_t denotes annual real GDP growth, $x_{t,q}$ are latent annualized quarterly growth rates, and e_t captures the approximation error. When quarterly GDP growth is observed, the latent state coincides with the data, that is $x_{t,q} = y_{t,q}$. Observed quarterly GDP growth is therefore treated as a noiseless measurement of the latent quarterly growth rate.

Equation (4) implies that annual GDP growth is a weighted average of quarterly growth rates spanning the current and previous calendar year. Since future realizations of annual GDP growth are not observed and our objective is to recover quarterly growth rates implied by annual SPF projections, we use the SPF consensus forecasts as observations for annual growth rates, i.e., y_t and y_{t+1} are replaced by $\text{SPF}_{t|t,q}^{FE}$ and $\text{SPF}_{t+1|t,q}^{FE}$.

When all but one quarterly growth rate entering equation (4) are observed, the remaining quarterly growth rate is uniquely determined. For example, at the end of calendar year t , when quarterly GDP growth is observed up to $y_{t,3}$, the SPF-consistent

quarterly growth rate for the fourth quarter of year t , $x_{t,4}$, can be recovered directly from equation (4).

However, in practice the SPF deadline typically precedes the release of the previous quarter's GDP. As a result, if more than one quarterly growth rate in equation (4) is unobserved, the implied quarterly growth rates are no longer identified. This issue is particularly relevant for h -quarters-ahead forecasts, which require exploiting both the current- and next-year SPF projections. While for the current year t some quarterly growth rates may already be available, annual SPF projections for year $t + 1$ imply that all quarterly growth rates for that year are unobserved in equation (4).

To resolve the identification issue, we impose structure on the dynamics of the latent state by modeling quarterly growth rates as a random walk:

$$x_{t,q} = Lx_{t,q} + v_{t,q}, \quad (5)$$

where L is the quarterly lag operator defined in equation (3), and $v_{t,q}$ denotes the random walk innovation.²

Together, equations (4) and (5) define a state-space system linking annual growth rates to latent quarterly dynamics. To implement this system on a quarterly frequency, we express the annual SPF projections as quarterly observations that enter the measurement equation only at the end of each calendar year. Specifically, we define annual growth rates from the fixed-event SPF projections as

$$y_{t+j,\tilde{q}}^A = \begin{cases} \text{SPF}_{t+j|t,q}^{FE} & \text{if } \tilde{q} = 4 \\ \text{missing} & \text{else,} \end{cases} \quad (6)$$

where $\tilde{q}$ denotes the quarter index of the quarterly state-space system, rather than the quarter in which the survey was conducted. Accordingly, in quarter q of year t , we use the

²[Hepenstick and Blunier \(2022\)](#) show that the random walk specification is robust to alternative dynamic specifications, such as an AR(2) process.

SPF projections for the current and next calendar year ($j = 0, 1$), and $y_{t+j,\tilde{q}}^A$ takes on the corresponding SPF projection when the system reaches the fourth quarter of the calendar year, while it is treated as missing in all other quarters.

For instance, in quarter 3 of year t , the available information consists of $\text{SPF}_{t|t,3}^{FE}$ and $\text{SPF}_{t+1|t,3}^{FE}$. These projections therefore enter the measurement equation as $y_{t,4}^A = \text{SPF}_{t|t,3}^{FE}$ and $y_{t+1,4}^A = \text{SPF}_{t+1|t,3}^{FE}$, while all other quarterly observations of $y_{t,\tilde{q}}^A$ are missing.

For observed quarterly GDP growth, no timing adjustment is required, since quarterly realizations enter the system contemporaneously. Accordingly, for the quarterly growth rates, q and $\tilde{q}$ coincide, and we define

$$y_{t,q}^Q = \begin{cases} y_{t,q} & \text{if observed} \\ \text{missing} & \text{else.} \end{cases} \quad (7)$$

Collecting the annual and quarterly measurement equations and the transition equation for the latent quarterly growth rates, the model can be written in state-space form as

$$\mathbf{y}_{t,q} = \mathbf{C}\mathbf{x}_{t,q} + \mathbf{u}_{t,q} \quad (8)$$

$$\mathbf{x}_{t,q} = \mathbf{A}L\mathbf{x}_{t,q} + \mathbf{w}_{t,q}, \quad (9)$$

where

$$\mathbf{y}_{t,q} = \begin{bmatrix} y_{t+j,\tilde{q}}^A \\ y_{t,q}^Q \end{bmatrix}, \quad \mathbf{x}_{t,q} = \begin{bmatrix} x_{t,q} \\ Lx_{t,q} \\ \vdots \\ L^6x_{t,q} \end{bmatrix}, \quad \mathbf{u}_{t,q} = \begin{bmatrix} e_{t,q} \\ 0 \end{bmatrix}, \quad \mathbf{w}_{t,q} = \begin{bmatrix} v_{t,q} \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

and $\mathbf{x}_{t,q}$ denotes the state vector. Here, L^k denotes the quarterly lag operator of order k , acting elementwise on the latent state vector, and $e_{t,q}$ captures the approximation error associated with the annual aggregation using information available up to quarter q in year t . Missing observations in $\mathbf{y}_{t,q}$ are handled in the standard way by the Kalman filter,

i.e., by conditioning on the available elements of the measurement vector at each quarter.

The coefficient matrices are defined as

$$\mathbf{C} = \begin{bmatrix} \frac{1}{16} & \frac{2}{16} & \frac{3}{16} & \frac{4}{16} & \frac{3}{16} & \frac{2}{16} & \frac{1}{16} \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & \mathbf{0}_{1 \times 6} \\ \mathbf{I}_6 & \mathbf{0}_{6 \times 1} \end{bmatrix},$$

where $\mathbf{0}$ are vectors of zeros, and $\mathbf{I}$ is the identity matrix.

4.2 Estimation and construction of quarterly SPF forecasts

Given the state-space representation in equations (8) and (9), the model could in principle be estimated by QML using the Kalman filter, jointly estimating the variance of the measurement error and the variance of the random walk innovation. In practice, however, the approximation error $e_{t,q}$ may be small, and estimating both variance components leads to weak identification and boundary solutions. This is a well-known pile-up problem in state-space models with persistent latent states, where QML estimators of variance parameters may exhibit probability mass at zero ([Stock and Watson, 1998](#)).

We therefore follow [Hepenstrick and Blunier \(2022\)](#) and calibrate the variance of the approximation error $e_{t,q}$ to the data. Alternatively, [Stock and Watson \(1998\)](#) suggest normalizing the variance of the measurement error in a first step. In our application, the results are robust to alternative calibrations of the approximation error variance, indicating that the resulting quarterly SPF-consistent forecasts are driven mainly by the annual aggregation constraint and the assumed state dynamics.

Given the calibrated measurement error variance, only the variance of the random walk innovation $v_{t,q}$ remains to be estimated by QML. Estimation uses the real-time GDP vintage available at each SPF survey deadline, as described in Section 3, ensuring that the filtering and smoothing procedure respects the information set available to survey respondents. Using later revised data would otherwise force the filter to reconcile GDP realizations that were not known at the time of the survey with the annual SPF projections reported in that survey round.

Conditional on the estimated parameters, the Kalman smoother delivers estimates of the latent quarterly growth rates from the current quarter through the fourth quarter of year $t + 1$. We interpret the smoothed latent states for horizons $h = 0, 1, \dots, 4$ as real-time quarterly SPF-consistent forecasts, i.e., implied quarterly forecasts $\text{SPF}_{t,q+h|t,q}$. The same filtering procedure can be applied to each survey respondent individually, yielding a panel of SPF-compatible forecasts, $\text{SPF}_{t,q+h|t,q}^i$.

5 Benchmark models and forecast evaluation

We assess the forecasting performance of the implied quarterly SPF forecasts for $h = 0, 1, \dots, 4$ by comparing their root mean squared forecast error (RMSFE) to common benchmark models (e.g., [Stark, 2010](#)). The benchmark models are estimated recursively using an expanding window of real-time vintages of real GDP growth. The first available vintage corresponds to the first quarter of 2001 and contains data starting in 1991. Because the SPF is released around mid-quarter, the first estimate of GDP for the previous quarter is typically published before the survey release. However, exact release dates for monthly GDP vintages are only available from 2014 onward. We therefore adopt a conservative approach and use the GDP vintage available in the second month of each quarter when estimating the benchmark models. As a result, the benchmark forecasts are based on GDP growth up to quarter $t - 1$, whereas the filtered SPF-consistent projections rely only on information up to quarter $t - 2$.

The autoregressive models of lag order p comprise direct and indirect autoregressions ($\text{AR}(p)$). The h -quarter-ahead forecasts of the direct $\text{AR}(p)$ model, i.e., $\text{DAR}(p)$, are estimated as

$$y_{t,q+h} = \alpha + \sum_{k=1}^p \beta_k^h L^k y_{t,q-1} + \varepsilon_{t,q}, \quad h = 0, \dots, 4,$$

where the latest observed growth rate is $y_{t,q-1}$. Estimating this model by OLS for each forecast horizon separately, we obtain the forecasts $\hat{y}_{t,q+h}^{\text{DAR}}$. Forecasts of the indirect AR

(IAR(p)) are produced by estimating an AR(p) model

$$y_{t,q} = \tilde{\alpha} + \sum_{k=1}^p \tilde{\beta}_k L^k y_{t,q-1} + \tilde{\varepsilon}_{t,q},$$

and iterating forward, resulting in $\hat{y}_{t,q+h}^{\text{IAR}}$. For both specifications, the lag length is selected recursively using the Bayesian information criterion.

In addition to the autoregressions, we consider a sample mean forecast (SM), computed as the average of all available quarterly growth rates in the vintage at time (t, q) :

$$\hat{y}_{t,q+h}^{\text{SM}} = \frac{1}{T_{t,q-1}} \sum_{k=1}^{T_{t,q-1}} L^k y_{t,q-1}, \quad h = 0, \dots, 4,$$

where $T_{t,q-1}$ denotes the number of observed growth rates included in that vintage, which contains data up to $y_{t,q-1}$. Finally, we consider the random walk (RW), or no-change forecast:

$$\hat{y}_{t,q+h}^{\text{RW}} = y_{t,q-1}, \quad h = 0, \dots, 4,$$

as simple benchmark models.

To construct actual outcomes of real GDP growth, we use the first releases of GDP:

$$y_{t,q}^{\text{act}} = 100 \left(\left(\frac{\text{GDP}_{t,q}^{(1)}}{\text{GDP}_{t,q-1}^{(*)}} \right)^4 - 1 \right),$$

where $\text{GDP}_{t,q}^{(1)}$ denotes the first release of quarter q in year t , and $\text{GDP}_{t,q-1}^{(*)}$ refers to the previous quarter value from the same vintage containing the first release. See Figure 1 for the timing of GDP releases and revisions.

To compare forecast performance, we report the RMSFE for each horizon $h = 0, 1, \dots, 4$ as

$$\text{RMSFE}_h = \sqrt{\frac{1}{T_{\text{eval}}} \sum_{t,q} \left(\hat{y}_{t,q+h} - y_{t,q+h}^{\text{act}} \right)^2},$$

where T_{eval} denotes the number of available evaluation periods. Forecasts $\hat{y}_{t,q+h}$ are either from the SPF-consistent Kalman procedure or one of the benchmark models (DAR, IAR, SM, RW). To formally evaluate forecast accuracy, we use the test of [Diebold and Mariano \(1995\)](#) and [West \(1996\)](#), and apply pair-wise model comparisons based on the mean squared forecast errors.

6 Empirical results

We first analyze the forecast performance and efficiency of the quarterly SPF-consistent forecasts in Section 6.1. Section 6.2 tests for deviations from rationality of the filtered SPF predictions. In Section 6.3, we study forecaster heterogeneity using a panel constructed from individual implied SPF forecasts. Finally, Section 6.4 extends the filtering procedure to incorporate annual fixed-horizon forecasts and complements the analysis with additional robustness checks.

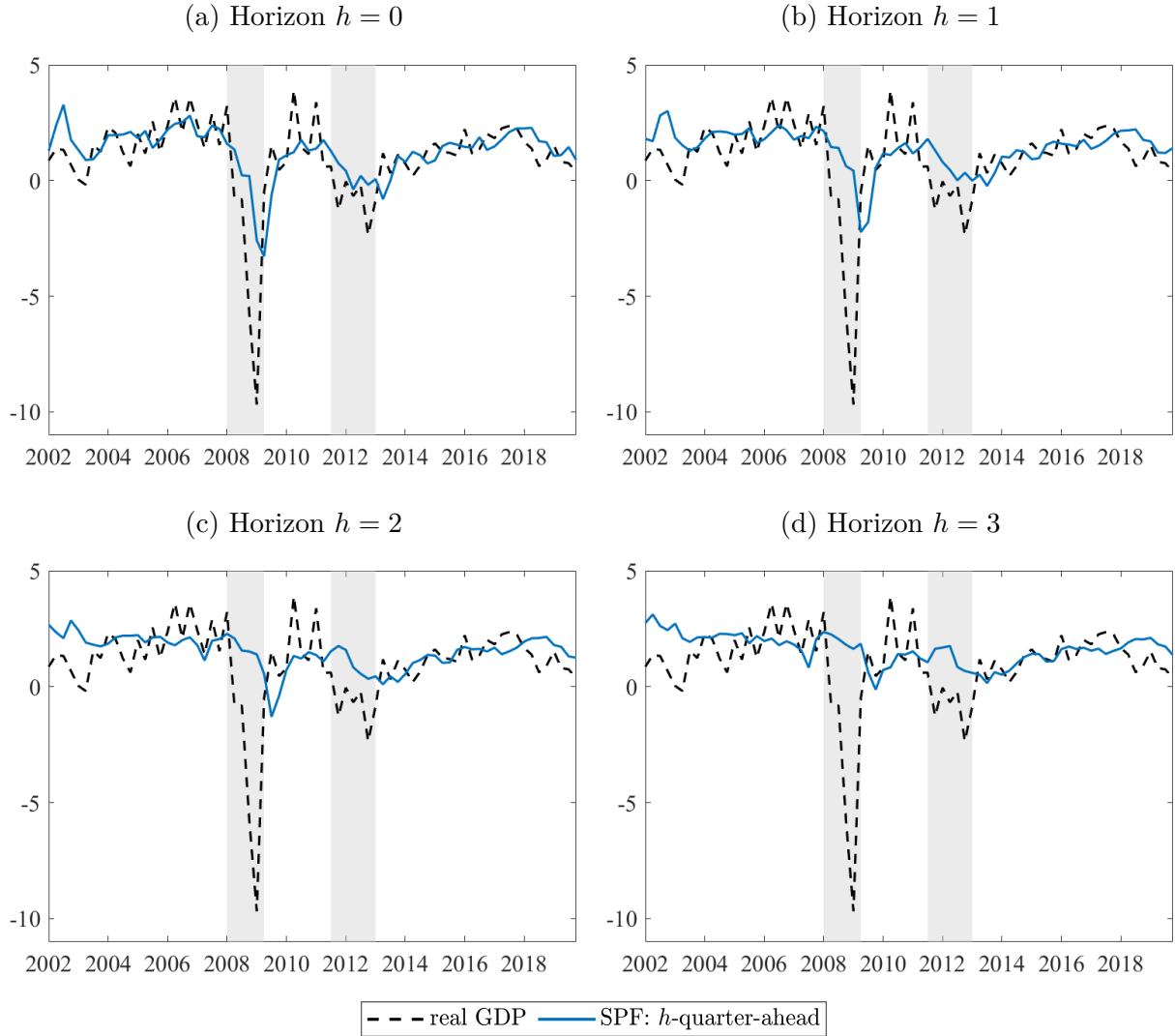
6.1 Consensus forecast performance

6.1.1 Descriptive evidence

Given the unprecedented nature of the COVID-19 shock, we analyze the pre- and post-pandemic periods separately. Excluding the pandemic, Figure 2 plots the real-time quarterly SPF forecasts filtered from annual fixed-event median projections against first releases of real GDP growth. At the shortest horizons, the 0- and 1-step-ahead quarterly forecasts track actual GDP growth reasonably well, although they underestimate the severity of the global financial crisis. For horizons of two and three quarters ahead, the SPF-consistent projections are smoother relative to realized GDP growth and show little variation across horizons.

Figure B.1 extends this comparison through the pandemic until the end of 2024. Clearly, at all horizons, the filtered forecasts cannot anticipate the pandemic and miss both the sharp downturn and the subsequent strong recovery in mid-2020. Across horizons, the

Figure 2: Quarterly implied SPF consensus forecasts



Notes: This figure plots the filtered quarterly real-time SPF forecasts consistent with the consensus SPF annual fixed-event forecasts against the first release of euro area real GDP growth. The sample period is 2002:Q1–2019:Q4. Vertical gray bands indicate euro area recessions as dated by the CEPR.

forecasts either smooth through the COVID-19 shock or adjust with some delay. Even in 2021, the implied SPF forecasts deviate strongly from realized GDP growth, and it takes until 2023 for forecasts and actual growth to closely align again across horizons.

Table A.1 reports summary statistics over the evaluation sample for first releases of real GDP growth and the implied quarterly SPF forecasts. Across all horizons, the SPF-consistent h -step-ahead forecasts are, on average, overly optimistic. In the pre-pandemic period, the standard deviation as well as the minimum and maximum of

the quarterly forecasts decrease with the forecast horizon, indicating convergence of the consensus forecasts at longer horizons. Moreover, the variation in the forecasts is considerably smaller than that of GDP growth. Extending the sample to include the COVID-19 pandemic increases the volatility of the filtered SPF forecasts. In this extended sample, the standard deviation of the SPF-consistent forecasts increases with h . However, it remains substantially lower than the volatility of GDP growth during that period.

6.1.2 Out-of-sample forecast evaluation

To formally evaluate the filtered SPF forecasts against benchmark models, Table 1 reports root mean squared forecast errors and presents results from pair-wise Diebold–Mariano tests comparing the SPF to the benchmarks. Since the first real-time vintage of GDP is available for 2001, the evaluation sample starts in 2002, which is the first year for which real-time forecasts are available for all horizons $h = 0, \dots, 4$.

Table 1: Out-of-sample forecast performance

Horizon	SPF	DAR(p)	IAR(p)	SM	RW
0	1.49	2.34*	2.43*	2.03*	2.32*
1	1.82	2.15**	2.55*	2.05*	2.50*
2	1.97	2.14*	2.62	2.08**	2.73*
3	2.06	2.20**	2.88	2.09	2.88*
4	2.13	2.17	2.84	2.10	2.91*

Notes: Quarterly real-time out-of-sample root mean squared forecast errors for horizons $h = 0, \dots, 4$, evaluated against first releases of annualized quarterly real GDP growth. SPF denotes the implied quarterly SPF forecasts based on annual median fixed-event projections. DAR(p) and IAR(p) are direct and indirect forecasts from AR(p) models. SM is the real-time sample mean forecast, and RW is the random walk (no-change) forecast. The evaluation period covers 2002:Q1–2019:Q4 (72 quarters). Stars indicate pair-wise one-sided Diebold–Mariano tests; *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Excluding the pandemic, the first column shows that the filtered SPF provides the most accurate forecasts at short horizons, namely for the current quarter and one quarter ahead. For horizons of two quarters and beyond, forecast performance deteriorates, consistent with earlier findings for the quarterly U.S. SPF (Stark, 2010). For horizons up to two quarters ahead, the SPF significantly outperforms the benchmark models. At longer horizons, only

the $\text{DAR}(p)$ and the random walk forecasts differ significantly from the SPF.

These findings are not driven by the global financial crisis. Panel (a) of Table A.2 excludes the year 2009. The resulting RMSFEs are generally smaller, and forecast errors increase with the horizon across models. However, the results confirm that the filtered SPF is highly competitive at short horizons and exhibits predictive performance similar to that of a simple backward-looking sample mean forecast at horizons of three quarters and beyond. Hence, the relative short-horizon advantage of the filtered SPF does not hinge on the exceptional volatility observed during 2009.

Lastly, Panel (b) of Table A.2 extends the sample until 2024. The COVID-19 pandemic, however, generates exceptionally large forecast errors at its onset in 2020 and during the subsequent year. This affects both the implied SPF and the benchmark models, rendering RMSFE comparisons difficult. We therefore omit the period 2020–2021 from the evaluation and find that the previous results continue to hold.

Overall, the results suggest that the information embedded in the annual fixed-event projections is particularly valuable at short horizons. Consistent with earlier evidence, the forecasts are informative about the near future but considerably less so for medium- and longer-term projections.

6.1.3 Bias and efficiency

Having established that the filtered SPF forecasts are competitive at short horizons, we next assess their bias and efficiency across forecast horizons. Bias is evaluated by comparing average realized GDP growth to the average forecast, while efficiency is examined using Mincer–Zarnowitz regressions:

$$y_{t,q+h}^{\text{act}} = \alpha_{MZ} + \beta_{MZ} \text{SPF}_{t,q+h|t,q} + \xi_{t,q+h|t,q},$$

where $\text{SPF}_{t,q+h|t,q}$ denotes the real-time filtered quarterly forecasts based on annual fixed-event SPF projections. Forecast efficiency implies the joint null hypothesis $H_0 : \alpha_{MZ} = 0$

and $\beta_{MZ} = 1$ (Mincer and Zarnowitz, 1969).

Table 2 reports the results for the evaluation period 2002–2019. The first column shows that forecasters are on average too optimistic about GDP growth across all horizons. The bias increases with the forecast horizon and becomes statistically significant at the 5% level for $h = 4$.

Table 2: Bias and efficiency of implied SPF consensus forecasts

Horizon	Bias	Mincer–Zarnowitz			
		α_{MZ}	β_{MZ}	p -value	R^2
0	-0.31* (0.19)	-0.49 (0.45)	1.14 (0.26)	0.17	0.41
1	-0.44* (0.25)	-0.18 (0.53)	0.81 (0.27)	0.11	0.14
2	-0.56* (0.31)	0.21 (0.45)	0.49 (0.24)	0.04	0.03
3	-0.67* (0.35)	0.58 (0.47)	0.23 (0.24)	0.00	0.01
4	-0.75** (0.37)	0.94 (0.51)	0.01 (0.33)	0.01	0.00

Notes: Bias and Mincer–Zarnowitz regressions for quarterly SPF forecasts at horizons $h = 0, \dots, 4$, evaluated against first releases of annualized quarterly real GDP growth. Implied quarterly SPF forecasts are filtered based on annual median fixed-event projections. Bias denotes the mean forecast error; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. The reported p -values refer to the joint null hypothesis $H_0 : \alpha_{MZ} = 0$ and $\beta_{MZ} = 1$. Newey–West standard errors are reported in parentheses. The evaluation period covers 2002:Q1–2019:Q4 (72 quarters).

The second and third columns report the Mincer–Zarnowitz regression estimates. At short horizons up to $h = 1$, the slope coefficient is close to unity, and the joint null of unbiasedness and efficiency cannot be rejected at conventional significance levels. Taken together, the bias and Mincer–Zarnowitz results suggest that forecasts at short horizons are broadly consistent with efficiency. For horizons of two quarters ahead and beyond, however, efficiency is rejected. The slope coefficients decline markedly and approach zero at longer horizons, indicating little systematic relationship between forecasts and subsequent realizations. Consistently, the R^2 decreases substantially from 0.41 at short horizons to

values close to zero.

Extending the sample to 2024 and excluding the pandemic years 2020–2021, Table A.3 shows that efficiency is also rejected at $h = 1$. Thus, while forecast performance remains strongest at short horizons, the evidence for efficiency becomes weaker in the extended sample. Overall, the filtered SPF forecasts remain informative at short horizons, while efficiency deteriorates as the forecast horizon increases.

6.2 Forecast adjustment and expectation dynamics

There is ample evidence that consensus forecasts of professional forecasters underreact to new information (Bordalo et al., 2020; Coibion and Gorodnichenko, 2015; Kohlhas and Walther, 2021). At the same time, Bordalo et al. (2020) document that, at the individual level, professional forecasters tend to overreact. For instance, Bürgi and Ortiz (2026) show that quarterly individual U.S. SPF forecasts display overreaction due to anchoring to their annual projections.

Since we apply a Kalman filter and smoother to obtain quarterly forecasts that are consistent with annual projections within a state-space framework, such behavioral biases are not straightforward to interpret as deviations from rational expectations. Nevertheless, it is informative to examine whether the implied quarterly SPF forecasts exhibit similar departures from rationality.

To test for deviations from rationality in the implied quarterly consensus forecasts, we follow Coibion and Gorodnichenko (2015)—henceforth CG—and regress forecast errors on forecast revisions at the consensus level for $h = 1, \dots, 3$:

$$y_{t,q+h} - \text{SPF}_{t,q+h|t,q} = \alpha_{CG} + \beta_{CG} FR_{t,q+h}^C + \xi_{t,q+h}, \quad (10)$$

where $y_{t,q+h} - \text{SPF}_{t,q+h|t,q}$ is the forecast error and $FR_{t,q+h}^C = \text{SPF}_{t,q+h|t,q} - \text{SPF}_{t,q+h|t,q-1}$ denotes the consensus forecast revision.

Applying the same test at the individual level, we pool the panel of implied quar-

terly forecasts $\text{SPF}_{t,q+h|t,q}^i$ and estimate a common intercept and slope coefficient across forecasters. Following [Bordalo et al. \(2020\)](#)—henceforth BGMS—we estimate:

$$y_{t,q+h} - \text{SPF}_{t,q+h|t,q}^i = \alpha_{BGMS} + \beta_{BGMS} FR_{t,q+h}^i + \xi_{t,q+h}^i, \quad (11)$$

where $FR_{t,q+h}^i$ denotes the forecast revision of forecaster i . In both equations (10) and (11), a positive slope coefficient indicates underreaction, and vice versa.

In addition, we follow [Bürge and Ortiz \(2026\)](#) and estimate revision autocorrelation (RA) regressions for the filtered consensus as well as individual forecasts:

$$FR_{t,q+h}^C = \alpha_{RA} + \beta_{RA} FR_{t,q+h-1}^C + \epsilon_{t,q+h}^C \quad (12)$$

$$FR_{t,q+h}^i = \alpha_{RA}^p + \beta_{RA}^p FR_{t,q+h-1}^i + \epsilon_{t,q+h}^i, \quad (13)$$

where the superscript p indicates coefficients estimated on the pooled panel of forecast revisions. Equations (12) and (13) are estimated for $h = 0, 1, 2$, as including lagged revisions further reduces the available forecast horizons. A positive coefficient β_{RA} indicates persistence in forecast revisions, i.e., the implied SPF forecasts exhibit underrevision in the sense that positive forecast revisions predict further positive revisions in the subsequent quarter.

In the following, we focus on the main evaluation period 2002–2019. Table 3 reports OLS results for the implied SPF consensus forecasts in Panel (a) and pooled unbalanced panel OLS regressions in Panel (b). Columns 1 to 4 present estimates from regressing forecast errors on forecast revisions according to equations (10) and (11). For the consensus forecasts, the SPF-consistent filtered predictions exhibit underreaction across all horizons. However, standard errors are sizable, and the coefficient is statistically significant only for $h = 0$ and decreases at longer horizons.

At the individual level, we likewise find underreaction for current-quarter and one-quarter-ahead forecasts. In contrast, for three quarters ahead, the coefficient estimate turns negative and significant. Consistent with BGMS, for $h = 3$ the individual implied SPF

Table 3: Forecast errors and revision dynamics of implied SPF forecasts

Panel (a): Consensus forecasts							
	Forecast error				Forecast revision		
	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 0$	$h = 1$	$h = 2$
$FR_{t,q+h}^C$	0.97** (0.48)	0.44 (0.48)	0.17 (0.50)	0.08 (0.57)	0.35*** (0.12)	0.41* (0.21)	0.16 (0.21)
Cons.	-0.19* (0.11)	-0.38* (0.20)	-0.54* (0.30)	-0.66** (0.32)	-0.08* (0.05)	-0.08** (0.04)	-0.10** (0.04)
R^2	0.15	0.02	0.00	0.00	0.09	0.11	0.02
Obs.	72	72	72	72	72	72	72

Panel (b): Individual forecasts							
	Forecast error				Forecast revision		
	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 0$	$h = 1$	$h = 2$
$FR_{t,q+h}^i$	0.53*** (0.06)	0.16** (0.07)	-0.05 (0.05)	-0.18*** (0.04)	0.16*** (0.03)	0.14*** (0.03)	-0.06 (0.02)
Cons.	-0.24*** (0.02)	-0.41*** (0.02)	-0.55*** (0.03)	-0.65*** (0.03)	-0.11*** (0.01)	-0.10*** (0.01)	-0.11*** (0.01)
R^2	0.06	0.00	0.00	0.00	0.02	0.02	0.00
Obs.	3,460	3,459	3,459	3,547	3,039	3,037	3,034

Notes: Forecast error regressions following CG and BGMS, and revision autocorrelation regressions. The left block reports estimates of equations (10) and (11), where forecast errors are regressed on forecast revisions. The right block reports estimates of equations (12) and (13), where forecast revisions are regressed on lagged forecast revisions. $FR_{t,q+h}^C$ and $FR_{t,q+h}^i$ denote the consensus and individual forecast revisions, respectively. Implied quarterly SPF forecasts are filtered based on annual median fixed-event projections. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Newey–West standard errors are reported in Panel (a) and cluster-robust standard errors in Panel (b). The evaluation period covers 2002:Q1–2019:Q4.

forecasts display overreaction to new information. Overall, the filtered quarterly forecasts behave broadly in line with recent evidence, showing underreaction at the consensus level and overreaction at the individual level at longer horizons.

Columns 5 to 7 in Table 3 show results for the autocorrelation regressions (12) and (13). For both the consensus and the panel of SPF-consistent quarterly forecasts, the forecast revisions are significantly positively autocorrelated at short horizons of up to one quarter ahead, indicating underrevision relative to rational expectations. This finding deviates from recent evidence for professional forecasters of the U.S. SPF, which tend to overrevise

their quarterly predictions (Bürge and Ortiz, 2026).

6.3 Forecaster heterogeneity

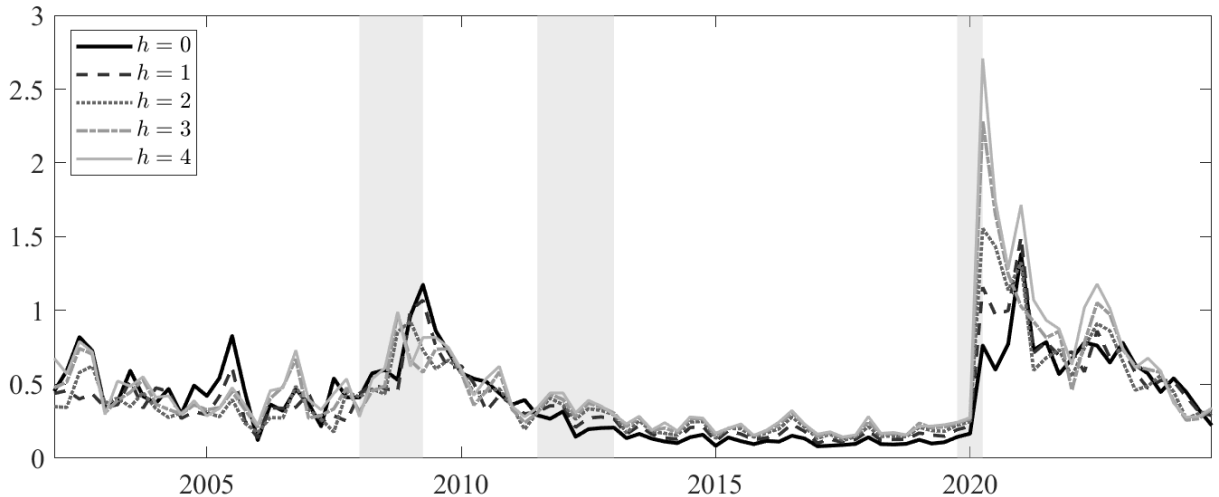
In this section, we examine heterogeneity across individual SPF-consistent predictions. For a survey on heterogeneous expectations among professional forecasters, see Conrad and Lahiri (2025). A common measure of disagreement is the interquartile range (IQR) of the cross-section of individual point forecasts (Abel et al., 2016; Glas, 2020). We construct the filtered h -step-ahead disagreement as

$$\text{SPF}_{t,q+h|t,q}^{\text{IQR}} = \text{SPF}_{t,q+h|t,q}^{0.75} - \text{SPF}_{t,q+h|t,q}^{0.25},$$

where $\text{SPF}_{t,q+h|t,q}^{0.25}$ and $\text{SPF}_{t,q+h|t,q}^{0.75}$ denote the 25th and 75th percentiles of the implied SPF forecasts.

Figure 3 plots the evolution of the resulting dispersion measure across forecast horizons. Each line represents disagreement at a specific forecast horizon at the time of the survey. There is notable time variation in the early 2000s, with a pronounced peak following the

Figure 3: Disagreement of quarterly implied SPF forecasts



Notes: This figure plots the interquartile range of implied SPF-consistent forecasts across survey dates. Disagreement is measured as the difference between the 75th and 25th percentiles of the cross-section of individual filtered SPF forecasts. Each line corresponds to a specific forecast horizon h . The sample period is 2002:Q1–2024:Q4. Vertical gray bands indicate euro area recessions as dated by the CEPR.

global financial crisis. After a period of relatively low dispersion across horizons in the aftermath of the financial crisis, disagreement spiked again in the second quarter of 2020 due to the COVID-19 pandemic. Interestingly, following the recession associated with the global financial crisis, disagreement is highest at the shortest forecast horizon. In contrast, during the pandemic, disagreement is largest for the four-quarter-ahead forecasts.

Table 4 reports descriptive statistics for the interquartile range. Excluding the pandemic period, Panel (a) shows that average dispersion increases with the forecast horizon (Patton and Timmermann, 2010), while the standard deviation and the range do not vary systematically with h . Extending the sample to 2024 in Panel (b), average dispersion rises at all horizons and remains largest for $h = 4$. In addition, the variation in disagreement increases with the forecast horizon, driven by the COVID-19 pandemic. In fact, the longer the forecast horizon, the larger the spike in disagreement among filtered forecasts.

Table 4: Summary statistics for disagreement of implied SPF forecasts

Panel (a): 2002–2019				
Horizon	Mean	Std. dev.	Min	Max
$h = 0$	0.34	0.25	0.08	1.17
$h = 1$	0.32	0.19	0.10	1.07
$h = 2$	0.32	0.17	0.12	0.93
$h = 3$	0.35	0.18	0.14	0.98
$h = 4$	0.39	0.20	0.14	0.99

Panel (b): 2002–2024				
Horizon	Mean	Std. dev.	Min	Max
$h = 0$	0.40	0.27	0.08	1.38
$h = 1$	0.40	0.27	0.10	1.50
$h = 2$	0.41	0.28	0.12	1.56
$h = 3$	0.45	0.33	0.14	2.28
$h = 4$	0.50	0.39	0.14	2.71

Notes: Summary statistics for disagreement among individual h -step-ahead quarterly SPF-consistent forecasts. The implied quarterly forecasts are based on individual annual SPF fixed-event projections and are estimated using a Kalman filter with smoothing. Panel (a) covers 2002–2019, while Panel (b) extends to 2024. The sample sizes are 72 and 92 quarterly observations, respectively.

6.4 Robustness

We assess the robustness of our findings along three dimensions. First, we extend the state-space framework to incorporate SPF fixed-horizon forecasts in addition to fixed-event forecasts. Second, we consider alternative definitions of the SPF consensus. Third, we compare the forecast accuracy when using only current-year forecasts instead of current- and next-year forecasts.

Incorporating SPF fixed-horizon forecasts The year-on-year fixed-horizon forecasts of the SPF provide an additional observation equation. Appendix C details the corresponding extension of the state-space representation outlined in Section 4 and describes the estimation and filtering procedure for the SPF-consistent quarterly forecasts.

The resulting filtered SPF forecasts are almost identical to those implied by exploiting only the fixed-event forecasts. Only during the economic turmoil associated with the COVID-19 pandemic do temporary differences emerge. Figure B.2 compares the two approaches. From 2022 onward, they again yield virtually identical forecasts. In unreported results, forecast accuracy is not improved by incorporating fixed-horizon projections. This suggests that the additional structure implied by fixed-horizon forecasts does not add information for the quarterly filtering procedure.

In addition to extending the measurement equation, we also allow for an AR(1) transition in the state equation (see Appendix C). The resulting SPF-consistent forecasts are virtually identical to the baseline, and all empirical results remain unchanged.

Alternative definitions of SPF consensus Instead of the median, we can also use the mean of individual SPF fixed-horizon forecasts as consensus. Forecast accuracy is unaffected by this choice. Moreover, the consensus forecast can be constructed as the median of the cross-section of individual filtered SPF forecasts, which again yields almost identical results.

Current-year versus next-year SPF projections So far, the filtering approach has relied on both current- and next-year forecasts. However, obtaining short-term forecasts does not always require next-year projections. For example, in the first quarter of a calendar year, quarterly implied forecasts up to the fourth quarter can be obtained using only current-year projections. In contrast, four-quarter-ahead forecasts always require next-year projections. Unreported results indicate that restricting the filter to current-year forecasts whenever possible does not affect the implied SPF, consistent with [Hepenstick and Blunier \(2022\)](#).

7 Conclusion

This paper constructs quarterly fixed-horizon forecasts of GDP growth that are consistent with annual fixed-event forecasts from the ECB SPF. We use an approximation of annual projections by quarterly growth rates that can be cast into a state-space representation. Combined with a standard Kalman filter and smoother, this framework allows us to recover quarterly SPF-consistent forecasts.

Using real-time data for the euro area, we obtain implied quarterly consensus SPF forecasts without look-ahead bias. A realistic out-of-sample analysis reveals that the quarterly forecasts are accurate at short horizons and outperform standard time-series benchmark models up to around two quarters ahead. At longer horizons, however, the filtered SPF forecasts are less accurate, indicating that the survey provides informative forecasts particularly at short horizons.

Extending the methodology to the cross-section of individual survey respondents, we construct a panel of quarterly SPF-consistent fixed-horizon forecasts. For the consensus forecasts, we do not find strong evidence of bias at short horizons, while the filtered SPF forecasts exhibit underreaction at the shortest horizon. In contrast, individual SPF-consistent forecasts overreact at the three-quarters-ahead forecast horizon. These findings are broadly consistent with existing evidence on observed quarterly survey

expectations, lending credibility to the filter approach used in this paper.

The approach employed in this paper opens several avenues for future research. The implied quarterly SPF forecasts could be used as inputs in macroeconomic forecasting models to construct probabilistic forecasts, such as density forecasts or prediction intervals. The approach could also be extended to incorporate histogram forecasts from the SPF, allowing researchers to recover higher-frequency measures of survey uncertainty. Finally, the methodology could be applied to other SPF variables, such as inflation or unemployment, to construct quarterly survey-consistent expectations for a broader set of macroeconomic variables.

References

- Abel, J., R. Rich, J. Song, and J. Tracy (2016). The measurement and behavior of uncertainty: Evidence from the ECB survey of professional forecasters. *Journal of Applied Econometrics* 31(3), 533–550.
- Adams, P. A., T. Adrian, N. Boyarchenko, and D. Giannone (2021). Forecasting macroeconomic risks. *International Journal of Forecasting* 37(3), 1173–1191.
- Ang, A., G. Bekaert, and M. Wei (2007). Do macro variables, asset markets, or surveys forecast inflation better? *Journal of Monetary Economics* 54(4), 1163–1212.
- Aruoba, S. B. (2020). Term structures of inflation expectations and real interest rates. *Journal of Business & Economic Statistics* 38(3), 542–553.
- Bańbura, M., D. Giannone, M. Modugno, and L. Reichlin (2013). Now-casting and the real-time data flow. In *Handbook of Economic Forecasting*, Volume 2, pp. 195–237. Elsevier.
- Bańbura, M., F. Brenna, J. Paredes, and F. Ravazzolo (2021). Combining Bayesian VARs with survey density forecasts: Does it pay off?
- Bordalo, P., N. Gennaioli, Y. Ma, and A. Shleifer (2020, September). Overreaction in macroeconomic expectations. *American Economic Review* 110(9), 2748–82.
- Bürge, C. and J. L. Ortiz (2026). Overreaction through anchoring. *International Journal of Forecasting* 42(2), 512–526.
- Capistrán, C. and A. Timmermann (2009). Disagreement and biases in inflation expectations. *Journal of Money, Credit and Banking* 41(2/3), 365–396.
- Clark, T. E., G. Ganics, and E. Mertens (2025, 03). Constructing fan charts from the ragged edge of spf forecasts. *The Review of Economics and Statistics*, 1–45.

- Clements, M. P. (2022). Forecaster efficiency, accuracy, and disagreement: Evidence using individual-level survey data. *Journal of Money, Credit and Banking* 54(2-3), 537–568.
- Clements, M. P., R. W. Rich, and J. S. Tracy (2023). Surveys of professionals. In *Handbook of Economic Expectations*, pp. 71–106. Academic Press.
- Coibion, O. and Y. Gorodnichenko (2015, August). Information rigidity and the expectations formation process: A simple framework and new facts. *American Economic Review* 105(8), 2644–78.
- Conrad, C. and K. Lahiri (2025). Heterogeneous expectations among professional forecasters. In *The Routledge Handbook of Economic Expectations in Historical Perspective*, pp. 105–123. Routledge.
- Crump, R. K., S. Eusepi, E. Moench, and B. Preston (2023). The term structure of expectations. In R. Bachmann, G. Topa, and W. van der Klaauw (Eds.), *Handbook of Economic Expectations*, pp. 507–540. Academic Press.
- Diebold, F. X. and R. S. Mariano (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics* 13(3), 253–263.
- Dovern, J., U. Fritsche, and J. Slacalek (2012). Disagreement among forecasters in G7 countries. *The Review of Economics and Statistics* 94(4), 1081–1096.
- Elliott, G., I. Komunjer, and A. Timmermann (2008). Biases in macroeconomic forecasts: Irrationality or asymmetric loss? *Journal of the European Economic Association* 6(1), 122–157.
- Faust, J. and J. H. Wright (2013). Forecasting inflation. In *Handbook of Economic Forecasting*, Volume 2, pp. 2–56. Elsevier.
- Ganics, G., B. Rossi, and T. Sekhposyan (2024). From fixed-event to fixed-horizon density forecasts: Obtaining measures of multihorizon uncertainty from survey density forecasts. *Journal of Money, Credit and Banking* 56(7), 1675–1704.

- Giannone, D., L. Reichlin, and D. Small (2008). Nowcasting: The real-time informational content of macroeconomic data. *Journal of Monetary Economics* 55(4), 665–676.
- Glas, A. (2020). Five dimensions of the uncertainty–disagreement linkage. *International Journal of Forecasting* 36(2), 607–627.
- Grishchenko, O., S. Mouabbi, and J.-P. Renne (2019). Measuring inflation anchoring and uncertainty: A U.S. and euro area comparison. *Journal of Money, Credit and Banking* 51(5), 1053–1096.
- Heinisch, K. (2025). Step by step—a quarterly evaluation of EU commission’s GDP forecasts. *Journal of Forecasting* 44(3), 1026–1041.
- Hepenstrick, C. and J. Blunier (2022). What were they thinking? Estimating the quarterly forecasts underlying annual growth projections.
- Hubert, P. (2015). ECB projections as a tool for understanding policy decisions. *Journal of Forecasting* 34(7), 574–587.
- Juodis, A. and S. Kučinskas (2023). Quantifying noise in survey expectations. *Quantitative Economics* 14(2), 609–650.
- Knüppel, M. and A. L. Vladu (2025). Approximating fixed-horizon forecasts using fixed-event forecasts. *Journal of Applied Econometrics* 40(4), 359–379.
- Kohlhas, A. N. and A. Walther (2021, September). Asymmetric attention. *American Economic Review* 111(9), 2879–2925.
- Kozicki, S. and P. A. Tinsley (2012). Effective use of survey information in estimating the evolution of expected inflation. *Journal of Money, Credit and Banking* 44(1), 145–169.
- Krüger, F., T. E. Clark, and F. Ravazzolo (2017). Using entropic tilting to combine BVAR forecasts with external nowcasts. *Journal of Business & Economic Statistics* 35(3), 470–485.

- Krüger, F. and H. Plett (2024). Prediction intervals for economic fixed-event forecasts. *The Annals of Applied Statistics* 18(3), 2635–2655.
- Lahiri, K. and X. Sheng (2010). Measuring forecast uncertainty by disagreement: The missing link. *Journal of Applied Econometrics* 25(4), 514–538.
- Mertens, E. and J. M. Nason (2020). Inflation and professional forecast dynamics: An evaluation of stickiness, persistence, and volatility. *Quantitative Economics* 11(4), 1485–1520.
- Mincer, J. A. and V. Zarnowitz (1969, December). The evaluation of economic forecasts. In *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, NBER Chapters, pp. 3–46. National Bureau of Economic Research, Inc.
- Nordhaus, W. D. (1987). Forecasting efficiency: Concepts and applications. *The Review of Economics and Statistics* 69(4), 667–674.
- Patton, A. J. and A. Timmermann (2010). Why do forecasters disagree? Lessons from the term structure of cross-sectional dispersion. *Journal of Monetary Economics* 57(7), 803–820.
- Rich, R. and J. Tracy (2021). A closer look at the behavior of uncertainty and disagreement: Micro evidence from the euro area. *Journal of Money, Credit and Banking* 53(1), 233–253.
- Schick, M. (2024). Nowcasting growth-at-risk using the survey of professional forecasters. *Available at SSRN 4859937*.
- Schick, M. and A. Opschoor (2025). Multivariate economic tail risk and scenario analysis using the survey of professional forecasters.
- Siklos, P. L. (2013). Sources of disagreement in inflation forecasts: An international empirical investigation. *Journal of International Economics* 90(1), 218–231.

- Stark, T. (2010). Realistic evaluation of real-time forecasts in the survey of professional forecasters. *Research Rap Special Report* (May).
- Stock, J. H. and M. W. Watson (1998). Median unbiased estimation of coefficient variance in a time-varying parameter model. *Journal of the American Statistical Association* 93(441), 349–358.
- Tödter, K.-H. (2010). How useful is the carry-over effect for short-term economic forecasting? *Discussion Paper Series 1: Economic Studies* (2010, 21).
- West, K. D. (1996). Asymptotic inference about predictive ability. *Econometrica* 64(5), 1067–1084.
- Zarnowitz, V. and L. A. Lambros (1987). Consensus and uncertainty in economic prediction. *Journal of Political Economy* 95(3), 591–621.

Appendices

A Tables

Table A.1: Summary statistics

Panel (a): 2002–2019					
	Horizon	Mean	Std. dev.	Min	Max
Real GDP growth		0.95	1.90	-9.68	3.90
SPF h -step-ahead	$h = 0$	1.26	1.07	-3.27	3.28
	$h = 1$	1.39	0.87	-2.23	3.00
	$h = 2$	1.51	0.72	-1.29	2.86
	$h = 3$	1.62	0.64	-0.12	3.12
	$h = 4$	1.70	0.63	0.18	3.34
Panel (b): 2002–2024					
	Horizon	Mean	Std. dev.	Min	Max
Real GDP growth		1.07	8.10	-40.33	61.13
SPF h -step-ahead	$h = 0$	1.06	1.45	-6.11	3.29
	$h = 1$	1.27	1.40	-8.28	4.34
	$h = 2$	1.54	1.63	-8.07	10.61
	$h = 3$	1.81	1.62	-2.20	13.75
	$h = 4$	2.02	1.72	-0.41	14.79

Notes: Summary statistics for quarterly Euro Area real GDP growth and h -step-ahead quarterly SPF forecasts consistent with annual SPF projections. Real GDP growth is measured using first releases of annualized quarter-over-quarter growth rates. The implied quarterly SPF forecasts are based on annual median fixed-event forecasts and are estimated using a Kalman filter with smoothing. Panel (a) covers 2002–2019, while Panel (b) extends to 2024. The sample sizes are 72 and 92 quarterly observations, respectively.

Table A.2: Out-of-sample forecast performance for alternative samples

Panel (a): Sample period 2002–2019 (excluding 2009)					
Horizon	SPF	DAR(p)	IAR(p)	SM	RW
0	1.20	1.40**	1.44*	1.55***	1.47***
1	1.34	1.59*	1.56*	1.57***	1.74***
2	1.47	1.63**	2.17	1.59**	2.20*
3	1.59	1.71*	2.57	1.61	2.59
4	1.64	1.65	2.51	1.63	2.54*

Panel (b): Sample period 2002–2024 (excluding 2020–2021)					
Horizon	SPF	DAR(p)	IAR(p)	SM	RW
0	1.43	2.22*	2.31*	1.97**	2.18*
1	1.75	2.04*	2.41*	1.99**	2.53**
2	1.93	2.03	2.47	2.01	2.77***
3	2.06	2.09	2.71	2.03	2.92*
4	2.23	2.05	3.13*	2.04	3.05**

Notes: Quarterly real-time out-of-sample root mean squared forecast errors for horizons $h = 0, \dots, 4$, evaluated against first releases of annualized quarterly real GDP growth. SPF denotes the implied quarterly SPF forecasts based on annual median fixed-event projections. DAR(p) and IAR(p) are direct and indirect forecasts from AR(p) models. SM is the real-time sample mean forecast, and RW is the random walk (no-change) forecast. The evaluation period in Panel (a) covers 2002:Q1–2019:Q4, excluding the global financial crisis year 2009 (68 quarters). Panel (b) extends to 2024, excluding the years 2020–2021 due to COVID-19 (84 quarters). Stars indicate pair-wise one-sided Diebold–Mariano tests; *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

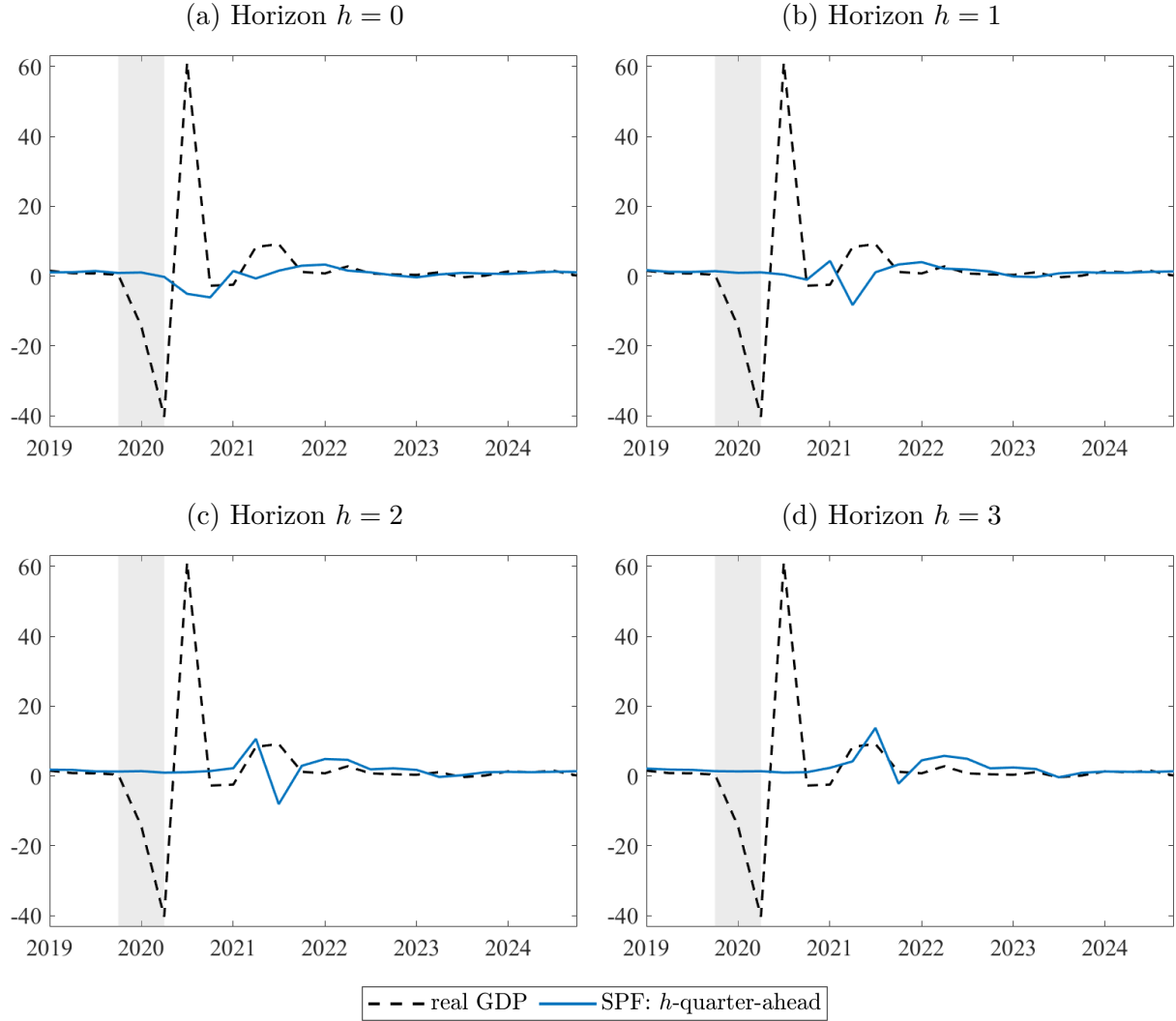
Table A.3: Bias and efficiency of implied SPF consensus forecasts for extended sample

Horizon	Bias	Mincer–Zarnowitz			
		α_{MZ}	β_{MZ}	p -value	R^2
0	-0.29* (0.16)	-0.35 (0.42)	1.05 (0.25)	0.11	0.38
1	-0.44* (0.22)	0.00 (0.44)	0.68 (0.22)	0.02	0.12
2	-0.61** (0.28)	0.35 (0.35)	0.38 (0.14)	0.00	0.03
3	-0.78** (0.32)	0.56 (0.36)	0.22 (0.10)	0.00	0.01
4	-0.93** (0.35)	0.83 (0.34)	0.06 (0.10)	0.00	0.00

Notes: Bias and Mincer–Zarnowitz regressions for quarterly SPF forecasts at horizons $h = 0, \dots, 4$, evaluated against first releases of annualized quarterly real GDP growth. Implied quarterly SPF forecasts are filtered based on annual median fixed-event projections. Bias denotes the mean forecast error; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. The reported p -values refer to the joint null hypothesis $H_0 : \alpha_{MZ} = 0$ and $\beta_{MZ} = 1$. Newey–West standard errors are reported in parentheses. The evaluation period covers 2002:Q1–2024:Q4, excluding the calendar years 2020 and 2021 (84 quarters).

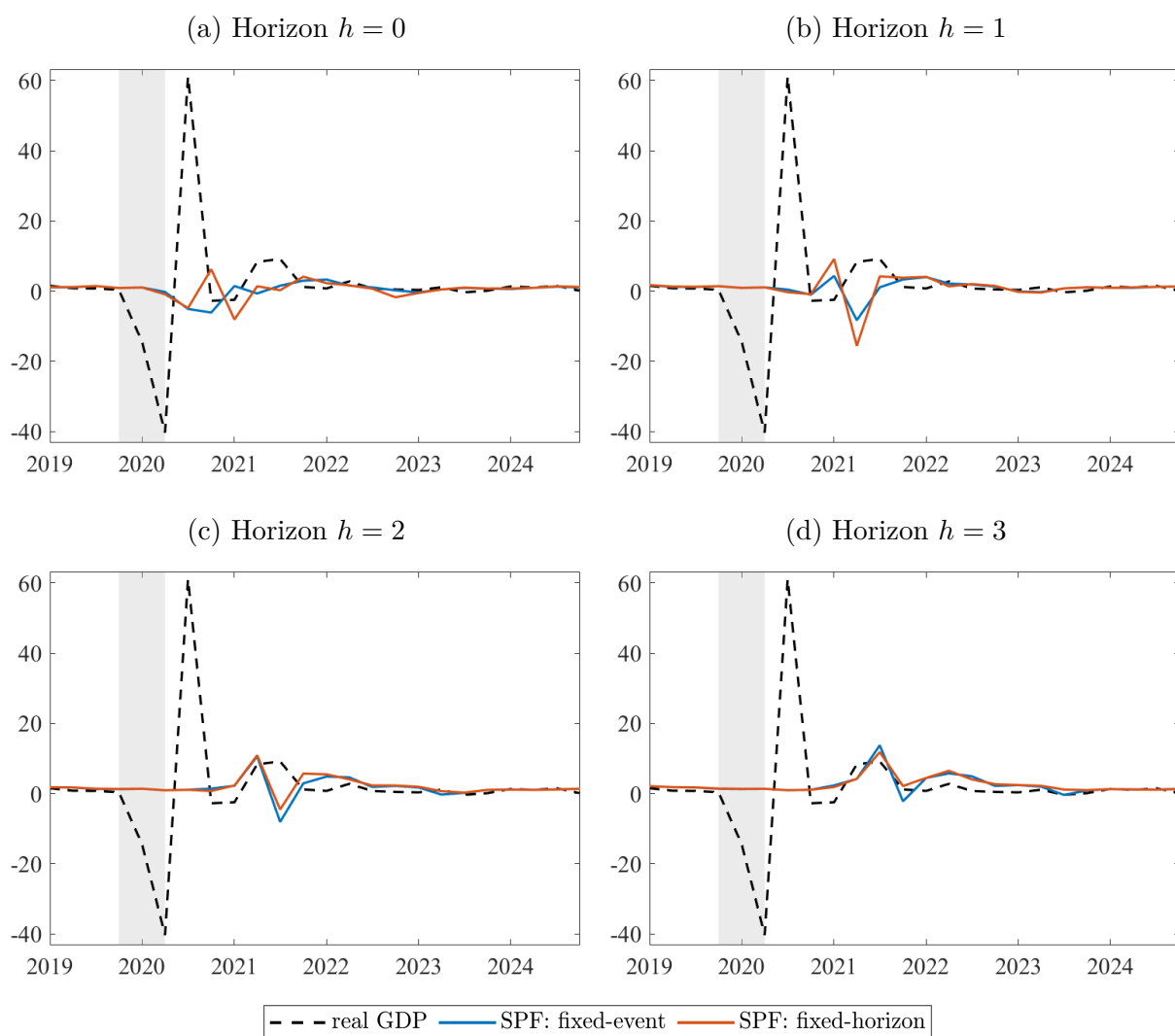
B Figures

Figure B.1: Quarterly implied SPF consensus forecasts for extended sample



Notes: This figure plots the filtered quarterly real-time SPF forecasts consistent with the consensus SPF annual fixed-event forecasts against the first release of euro area real GDP growth. The sample period is 2019:Q1–2024:Q4. Vertical gray bands indicate euro area recessions as dated by the CEPR.

Figure B.2: Quarterly implied SPF consensus forecasts including fixed-horizon projections



Notes: This figure plots the filtered quarterly real-time SPF forecasts against the first release of euro area real GDP growth. The blue line shows forecasts consistent with the consensus SPF annual fixed-event projections, while the red line combines both fixed-event and fixed-horizon projections. The sample period is 2019:Q1–2024:Q4. Vertical gray bands indicate euro area recessions as dated by the CEPR.

C Extensions to the state-space system

This appendix outlines extensions to the baseline state-space system discussed in Section 4, including modifications to the measurement and transition equations.

Incorporating SPF fixed-horizon forecasts Extending the state-space to incorporate the SPF fixed-horizon forecasts affects only the measurement equation. The year-on-year forecasts are approximated by

$$y_{t,q}^{YoY} = \frac{1}{4} \left(x_{t,q} + Lx_{t,q} + L^2x_{t,q} + L^3x_{t,q} \right) + \eta_{t,q}, \quad (\text{C.1})$$

where $\eta_{t,q}$ denotes the approximation error. Thus, in quarter q of year t , the yearly fixed-horizon forecast of the SPF enters the observation equation according to

$$y_{t,q+2}^{FH} = \begin{cases} \text{SPF}_{q+2|t,q}^{FH} & \text{if observed} \\ \text{missing} & \text{else.} \end{cases}$$

Adding the additional observation equation, the measurement equations utilizing both the fixed-event and fixed-horizon forecasts read

$$\mathbf{y}_{t,q}^{FH} = \begin{bmatrix} y_{t+j,\tilde{q}}^A \\ y_{t,q}^Q \\ y_{t,q}^{FH} \end{bmatrix}, \quad \mathbf{C}_{t,q}^{FH} = \begin{bmatrix} \frac{1}{16} & \frac{2}{16} & \frac{3}{16} & \frac{4}{16} & \frac{3}{16} & \frac{2}{16} & \frac{1}{16} \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ & & \mathbf{c}_{t,q}^{FH} & & & & \end{bmatrix}, \quad \mathbf{u}_{t,q}^{FH} = \begin{bmatrix} e_{t,q} \\ 0 \\ \eta_{t,q} \end{bmatrix}.$$

The vector $\mathbf{c}_{t,q}^{FH}$ contains four consecutive entries equal to $1/4$ and zeros elsewhere, reflecting the approximation to the four-quarter average growth rate in equation (C.1). The position of the first non-zero element depends on the target quarter of the SPF fixed-horizon forecast. For example, if $\text{SPF}_{q+2|t,q}^{FH}$ targets the second quarter, the first non-zero entry is the third element. Both approximation errors are calibrated to the data, and the variance of the random-walk innovations is estimated by QML.

Allowing for AR(1) dynamics in the transition equation Instead of a random-walk specification, we can generalize the transition equation to allow for AR(1) dynamics:

$$x_{t,q} = \phi_0 + \phi_1 Lx_{t,q} + v_{t,q},$$

with constant and slope coefficients ϕ_0 and ϕ_1 , respectively. In matrix form, the transition equation becomes

$$\mathbf{x}_{t,q} = \boldsymbol{\phi}_0 + \mathbf{A}L\mathbf{x}_{t,q} + \mathbf{w}_{t,q},$$

where $\boldsymbol{\phi}_0$ is the constant and $\mathbf{A}$ the coefficient matrix

$$\boldsymbol{\phi}_0 = \begin{bmatrix} \phi_0 \\ \mathbf{0}_{6 \times 1} \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} \phi_1 & \mathbf{0}_{1 \times 6} \\ \mathbf{I}_6 & \mathbf{0}_{6 \times 1} \end{bmatrix}.$$

The AR(1) coefficients ϕ_0 and ϕ_1 are estimated jointly with the innovation variance of the transition equation by QML.